ATU VIRTUAL LABORATORY

Regenerative and Friction Brake-Force Allocation: learning guide

Wheel-driven hub generation, disc friction braking and editable calibration

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A practical guide from first use to test design. The system is a browser-based educational simulation; physical hub-motor and vehicle stopping validation are separate activities.

Your first experiment

Keep the baseline values and select Run Model. Read requested force, regenerative force and friction force. Their relationship is simple: regeneration plus friction must equal requested force. At baseline, demand 0.4, mass 1500 kg and maximum service deceleration 3.5 m/s² request 2100 N. All 2100 N are regenerative at the default operating point; friction is 0 N. Battery charging power is 30.80 kW and charging current is 88.00 A.

Now set Battery SOC to 95%. At the default calibration the battery no longer accepts regeneration, so regenerative force becomes 0 N and friction force becomes 2100 N. This is an easy first verification: the allocation changes but the total demand remains satisfied. Restore defaults before the next test.

Understand the hub and brake disc

The road turns the tyre and wheel. During regenerative braking the wheel back-drives the hub motor, which acts as a generator and resists rotation. Electrical recovery still needs battery permission and adequate motor capability. The fixed stator stays still; the rotor turns. Choose Generator cutaway to hide the disc face and see the rotor; use Brake-disc close-up to inspect the disc and fixed caliper.

The brake disc turns with the wheel. The caliper stays fixed and presses friction pads against the rotating disc. Friction supplies braking that regeneration cannot provide. The colour cue represents a braking share, not a measured disc temperature.

One representative wheel hub illustrates the mechanism. Numerical forces remain vehicle totals; this model does not split braking among axles or individual wheels. The view is a generic geared hub cutaway, not manufacturer CAD. The reference ratio is 9:1. The Direct-drive hub button sets it to 1:1, recalculates the allocation and makes wheel and rotor angular speeds equal. Keep the chosen ratio in every test report. No hydraulic pressure, tyre grip, ABS, ESC or detailed planetary gear dynamics are solved.

Time, step time and playback in plain language

A calculation step is how often a time-based solver updates its equations. An input step time is when an input suddenly changes, such as waiting 20 simulated seconds before applying a new temperature. These are different meanings of step.

This particular module calculates one operating point. It has no integration time step or timed input-step parameter. Changing an input recalculates that point; it does not create a driving history. The minus/plus input buttons change a number by a small amount, not by advancing model time.

Run Model plays an illustrative stop using the requested constant deceleration. At baseline the estimated stop takes 11.90 physical seconds and 99.21 m. It is scaled to 8 screen seconds so it is easier to watch. The

progress time is viewing time. Pause freezes the picture; Resume continues it.

The hub angles use estimated travelled distance divided by the active wheel radius. Generator angle is wheel angle times the selected gear ratio. Numerical operating-point results remain fixed during playback. SOC, battery temperature and recovered energy do not evolve. High-speed rotation may strobe; use numeric RPM for measurements.

What the inputs mean

Brake demand is a fraction: 0 means no braking and 1 requests the selected maximum service deceleration. Requested deceleration equals demand times that maximum. Requested force equals mass times requested deceleration.

Vehicle speed sets wheel speed, motor speed, speed fade and the power-to-force conversions. Battery SOC and temperature describe charge acceptance at the selected operating point. Voltage converts a charging-current ceiling into a competing power ceiling.

Motor torque, motor power, battery current and battery power are separate ceilings. Increasing one ceiling may do nothing when another remains smaller. Regeneration Enabled removes or permits the regenerative contribution. Turning it off sends the requested braking work to friction.

Descriptions showing numerical thresholds refer to reference defaults. Your edited Advanced calibration values are the active assumptions; exported JSON records them. Suggested input ranges are guidance. Impossible inputs such as non-positive mass or voltage are rejected.

Read the charts and verify the result

The allocation bar compares regenerative and friction shares. Green means regeneration; amber means friction. At tiny demand below the active force-protection threshold, protected percentages may not total 100%. Check force values instead.

The force chart compares demand with the motor torque, motor power and battery force ceilings. The smallest candidate is highlighted before the enable/availability gates. The active-limit banner explains the final restriction.

The factor chart shows speed, SOC and temperature multipliers. Zero blocks that path; one means full availability under that factor. The power chart separates mechanical regenerative power from power accepted by the battery. Conversion efficiency makes these different.

Force-balance residual equals demand minus regeneration minus friction. A value near zero verifies the allocation arithmetic. It does not prove actual road stopping performance, tyre adhesion or hardware safety.

Edit advanced calibration confidently

Open Advanced calibration and type a new value. With Live Update checked, a valid edit immediately updates the preview. With it unchecked, select Run Model to apply the complete experiment. Any edit stops playback. Invalid settings keep the last valid readings and block Run and Export until corrected.

Radius, maximum deceleration, gear ratio and numerical protections must be positive. Efficiencies must be above zero and at most one. Speed cut-off must be below full regeneration speed. SOC thresholds must be ordered within 0-100%. Cold/hot thresholds must remain in ascending order.

Reset calibration defaults restores calibration only. Reset to Baseline restores this module's operating inputs and calibration. The shared Master reset to defaults reloads the module from its standard starting state, clears

the current run and discards unsaved edits. Export first. Saved external records or lecturer questions are not deleted.

Advanced settings are editable model assumptions, not measured production calibration. A larger experiment scope does not remove the consistency checks required to keep the equations meaningful.

Worked tests, from basic to advanced

Test 1: baseline. Expect 2100 N demand, 2100 N regeneration, 0 N friction, 30.80 kW battery charging and 88.00 A. Check the sum and export the JSON.

Test 2: SOC stop. Set SOC to 95% with default calibration. Expect 0 N regeneration and 2100 N friction. Then lower SOC to 50% and verify recovery returns.

Test 3: direct-drive hub. Set gear ratio to 1. The default motor torque ceiling corresponds to about 534.19 N at the wheel, so friction supplies about 1565.81 N. Inspect the lower motor RPM and equal wheel/rotor angular speeds. Restore ratio 9 to compare.

Test 4: change the acceptance threshold. With SOC at 95%, set SOC stop to 98 while derating starts at 80. Regeneration can return at a reduced acceptance factor. Explain why this differs from the default 95% cut-off. Export the changed calibration with your result.

Test 5: invalid calibration. Enter efficiency 1.2 or put full-regeneration speed below the cut-off. Run and Export should be blocked; the model must not emit invalid numbers. Restore calibration defaults and verify the baseline again.

Test 6: no braking. Set demand to zero at nonzero speed. Both braking forces are zero. The reference stopping-time zero is a sentinel for no finite stopping estimate, not an instantaneous stop. Playback remains off.

A student test report

Record the question, expected effect, all operating inputs and all edited calibration values. Change one variable at a time, identify the active restriction and compare before/after numerical results.

Keep force balance, motor RPM, acceptance factors and battery power/current in the evidence. Describe the direction wheel to generator during regeneration and disc to friction braking. Export JSON, which retains raw inputs, active calibration, results, warnings and playback state.

Separate calculation results from animation cues. This implementation follows the supplied BrakeLab programmer guide and canonical prompt; native SLX traces, physical hub-motor measurements and real stopping tests were not supplied as validation.

Reference equations and evidence

Demand force = mass × demand fraction × maximum service deceleration. Wheel speed = vehicle speed / wheel radius. Motor speed = wheel speed × gear ratio. Motor-torque force ceiling = motor torque × gear ratio × drivetrain efficiency × speed factor / wheel radius.

Battery power allowance combines current-derived and direct power ceilings with separate SOC and temperature factors. Regeneration receives the permitted minimum of demand and the available force ceilings. Friction supplies the remainder. Battery current = accepted electrical power / battery voltage.

The original pure allocation equations and baseline contract remain the reference; the editable controller passes your calibration to them. The additional 3D view is a kinematic illustration, not a new dynamic vehicle or thermal solver. Source documents remain in docs/reference/brakelab and the supplied project documentation.

Operating input glossary

Brake demand (ratio). Fraction of the 3.5 m/s² maximum service deceleration. The calculation normalises demand to 0-1 and warns about raw out-of-range entries.

Vehicle mass (kg). Mass multiplies requested deceleration to produce the required braking force. It must be positive.

Vehicle speed (km/h). Sets wheel/motor speed, low-speed fade and power-derived force limits. Negative entries are calculated as zero with a warning.

Battery SOC (%). State of charge is an independent input, not an evolving state. Acceptance derates above 80% and reaches zero at 95%.

Battery temperature (°C). Full acceptance is 15-40 °C. It falls to zero at –5 °C and 55 °C. This model does not simulate battery heating.

Battery voltage (V). Converts the charging-current limit into power and calculated battery power into current. It must be positive.

Charge-current limit (A). Maximum battery charging current; voltage × current gives the competing charging-power limit. Zero prevents recovery.

Charge-power limit (kW). Direct charging-power ceiling before SOC and temperature derating. The smaller of this and voltage × current controls base acceptance.

Motor regenerative-torque limit (N·m). Motor torque is multiplied by gear ratio, drivetrain efficiency and low-speed factor, then divided by wheel radius to obtain a force limit.

Motor regenerative-power limit (kW). Motor power is converted to a regenerative-force limit using speed. Friction braking supplies any remaining requested force.

Regeneration Enabled (). Off sends all requested braking force to friction. The reusable function also accepts a finite numeric enable factor, clamped to 0-1, as in the reference.

Advanced calibration glossary

Maximum service deceleration | Default 3.5. The strongest requested slowdown at brake demand 1. A value of 3.5 means speed falls by 3.5 metres per second each second under the illustrative constant-deceleration assumption.

Wheel radius | Default 0.31. Distance from axle centre to the tyre contact radius, metres. A larger radius needs more torque for the same braking force. It also changes wheel speed at a given road speed.

Hub gear ratio | Default 9. Generator speed divided by wheel speed. Ratio 9 means nine motor revolutions per wheel revolution. Ratio 1 represents a direct-drive hub. This value also changes the motor torque-to-wheel-force conversion.

Drivetrain efficiency | Default 0.92. Fraction retained in the source motor-to-wheel force relationship: greater than zero, at most one. The supplied reference relationship is retained for regeneration.

Recovery efficiency | Default 0.88. Fraction of mechanical regenerative power reaching the battery: greater than zero, at most one. The remainder is not recovered electrical power.

Regeneration speed cut-off | Default 2. At or below this road speed, regeneration availability is zero. The value is in metres per second; multiply by 3.6 for km/h.

Full-regeneration speed | Default 5. Above this speed the speed factor is one. Between cut-off and this value, it ramps gradually from zero to one. Must exceed cut-off.

SOC derating threshold | Default 80. Battery acceptance starts decreasing above this state of charge. Must be below the SOC stop threshold.

SOC stop threshold | Default 95. Battery acceptance reaches zero at this SOC. Friction supplies the requested braking force. Thresholds must be ordered within 0 to 100%.

Cold cut-off | Default -5. At or below this battery temperature, charge acceptance is zero.

Cold full-acceptance threshold | Default 15. Charge acceptance reaches its full temperature factor here. Must exceed cold cut-off.

Hot full-acceptance threshold | Default 40. Above this temperature, charge acceptance starts decreasing. Must not be below cold full acceptance.

Hot cut-off | Default 55. At or above this temperature, charge acceptance is zero. Must exceed the hot full threshold.

Minimum calculation speed | Default 0.1. Positive denominator protection for power-to-force division near zero speed. It is a numerical assumption, not a claim of useful regeneration at rest.

Small-force protection | Default 1. Protects percentage denominators and the no-braking diagnostic. Default 1 N. Below this threshold the displayed shares need not add to 100%.

Output glossary

Requested deceleration (m/s²). Requested slowdown: demand fraction times the selected maximum deceleration. Export: requestedDecelerationMps2

Requested force (N). Total requested wheel braking force: vehicle mass times requested deceleration. Export: requestedForceN

Requested wheel torque (N·m). Requested force times wheel radius; torque needed at the wheel. Export: requestedWheelTorqueNm

Operating-point speed (m/s). Input road speed converted from km/h to metres per second by dividing by 3.6. Export: vehicleSpeedMps

Wheel speed (rpm). Wheel rotational speed from road speed and the active radius. Export: wheelSpeedRpm

Motor speed (rpm). Generator rotational speed: wheel RPM times the active hub ratio. Export: motorSpeedRpm

Speed factor (ratio). Regeneration availability from speed, between zero and one. Export: speedFactor

SOC factor (ratio). Battery charge-acceptance factor from SOC, between zero and one. Export: socFactor

Temperature factor (ratio). Battery charge-acceptance factor from temperature, between zero and one. Export: temperatureFactor

Current-derived charge power (kW). Voltage times the charging-current ceiling, expressed in kW. Export: currentDerivedChargePowerKw

Base charge power (kW). Smaller of direct and current-derived charging-power ceilings, before acceptance factors. Export: baseChargePowerKw

Allowed battery power (kW). Battery charging-power allowance after the separate SOC and temperature factors. Export: allowedBatteryPowerKw

Torque force limit (N). Motor regenerative torque converted to wheel force through ratio, efficiency, radius and speed factor. Export: torqueForceLimitN

Power force limit (N). Motor regenerative power converted to a wheel-force ceiling using protected speed. Export: powerForceLimitN

Battery force limit (N). Battery charging-power allowance converted to a regenerative-force ceiling with recovery efficiency and speed. Export: batteryForceLimitN

Regenerative force (N). Part of braking supplied by the permitted wheel-driven generator path. Export: regenerativeForceN

Friction force (N). Requested force left after regeneration; supplied by the friction brakes. Export: frictionForceN

Regenerative wheel torque (N·m). Regenerative force times wheel radius. Export: regenerativeWheelTorqueNm

Friction wheel torque (N·m). Friction force times wheel radius. Export: frictionWheelTorqueNm

Mechanical regenerative power (kW). Mechanical power removed by regeneration: regenerative force times road speed, in kW. Export: mechanicalRegenPowerKw

Battery charging power (kW). Electrical power recovered into the battery after recovery efficiency. Export: batteryChargePowerKw

Regenerative current (A). Recovered electrical power divided by battery voltage. Export: regenerativeCurrentA

Regenerative share (%). Regenerative force as a percentage of demand using the protected denominator. Export: regenerativePercent

Friction share (%). Friction force as a percentage of demand using the protected denominator. Export: frictionPercent

Force-balance residual (N). Requested force minus regenerative force minus friction force; should be nearly zero. Export: forceBalanceResidualN